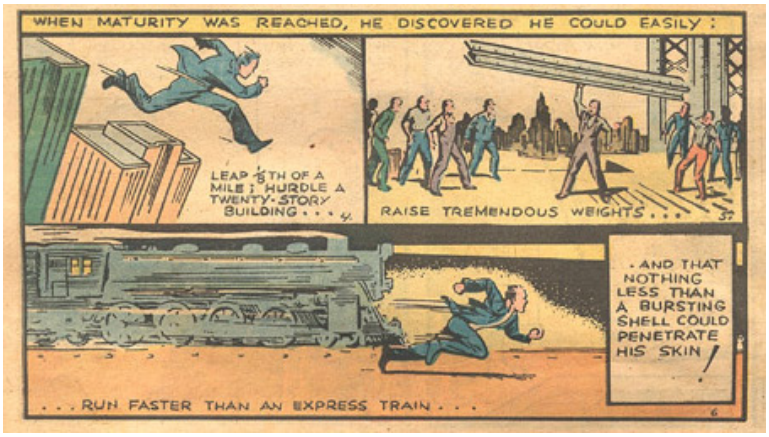
Dynamics: Problem Sheet 1 (of 8)

1. Superman (in his original incarnation) could hurdle a twenty-storey building. The source of his amazing strength was said to be the fact that his home planet of Krypton has a higher gravity than that of Earth, so that his muscles were necessarily stronger (rather like the way astronauts can leap large distances on the moon).





Consider the following model for a jump. Suppose that as Supermanjumps his legs provide a constant force F while he is contact with the ground. Suppose that when he crouches he lowers his centre of mass by 1 metre. Thus, if y is the height of his centre of mass in metres from the standing position, and m is his mass, he is acted on by a vertical force

$$f(y) = \begin{cases} F - mg & -1 < y \leq 0, \\ -mg & y > 0. \end{cases}$$

(a) Given that he starts at rest in the crouched position ($\dot{y}(0) = 0$, $y(0) = -1$), use Newton's Second Law to find his vertical velocity at the time his feet leave the ground (i.e. when $y = 0$);

(b) Show that he reaches his maximum height at a time

$$t = \sqrt{\frac{2}{(F/m - g)}} + \frac{\sqrt{2(F/m - g)}}{g},$$

and that this height is $y = F/mg - 1$.

(c) If a twenty-storey building is 100 m high, and Superman can jump only 1 m high on his home planet (since he is a "normal" man there), how much stronger is Krypton's gravity than Earth's?

2. A cannon at the origin O fires a shell with speed V at an angle α to the horizontal.

(a) Write down Newton's second law for the shell, with suitable initial conditions. Solve this differential equation to show that the trajectory of the shell is given by

$$x(t) \;=\; t\,V\cos\alpha\;, \qquad z(t) \;=\; -\frac{1}{2}gt^2 + t\,V\sin\alpha\;.$$

(b) Obtain the equation for the path of the shell, expressing the height z as a function of x.

(c) Suppose now that V is fixed but the angle α may be varied. Find the upper boundary curve $z = z(x)$ of the set of points in the (x, z) plane which it is possible to hit with a shell.

3. (a) Find the dimensions of kinetic energy ($\frac{1}{2}m|\dot{\mathbf{r}}|^2$) and linear momentum ($\mathbf{p} = m\dot{\mathbf{r}}$) in terms of the fundamental dimensions L, M and T.

(b) (The atomic bomb): An essentially instantaneous release of an amount of energy E from a very small volume creates a rapidly expanding high pressure fireball, bounded by a very strong thin spherical shock wave across which the pressure drops abruptly to atmospheric. The pressure inside the fireball is so great that the ambient atmospheric pressure is negligible by comparison, and the only property of the air that determines the radius $r(t)$ of the fireball is its density ρ (the mass of air per unit volume). Show dimensionally, by identifying the only possible combination of E, t and ρ, that

$$r(t) \quad \sim \quad E^{1/5}\, t^{2/5}\, \rho^{-1/5} \ .$$

This result, due to G. I. Taylor, can be used to deduce E from observations of $r(t)$. Taylor's publication of this apparently caused considerable embarrassment in US military scientific circles, where it was regarded as top secret.

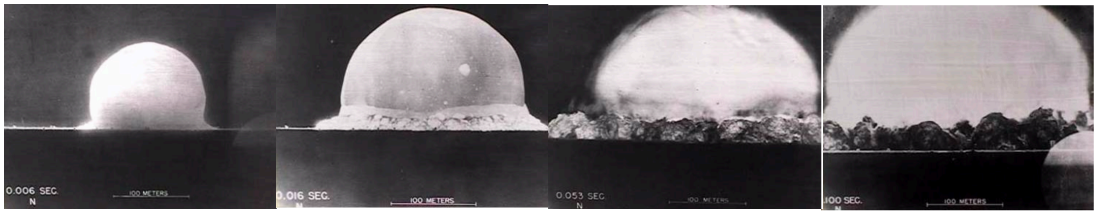


Figure 1: The published images off which Taylor based his calculation.

(c) (optional) Noting from the images above that the radius was approximately 80 meters at time $t = 0.006$ seconds, and using $\rho = 1.2$ kg/m^3 as the density of the surrounding air, estimate the energy E in the shock wave.